Essential Summary

A Generalized Padé–Lindstedt–Poincaré Method for Predicting Homoclinic and Heteroclinic Bifurcations of Strongly Nonlinear Autonomous Oscillators

Zhenbo Li, Jiashi Tang

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1 Motivation and Problem Statement

Two major classes of methods exist for analyzing homoclinic/heteroclinic (H/H) orbits of the oscillator $\ddot{x} + g(x) = \varepsilon f(x, \dot{x})$:

- **Perturbation-based methods** (HP, HLP, elliptic L–P, GHFP): accurate but require exact analytic solution of the unperturbed system, and involve **heavy derivation and integration** at each order. For high-order polynomial $g(x)$, these become intractable.
- **Padé-based methods**: no integration needed, but only work for $\varepsilon = 0$ (conservative limit). Cannot handle the damped case $\varepsilon \neq 0$.

This work proposes the **generalized Padé–Lindstedt–Poincaré (GPLP) method**: a synthesis that inherits the systematic perturbation framework of the L–P method but **replaces every integration step with generalized Padé approximation**. The result: a perturbation method that requires essentially no integration—each-order perturbation equation is solved by constructing a tailored generalized Padé approximant and matching coefficients.

2 The Generalized Padé Approximation (GPA)

The authors first generalize the classical Padé definition. In the GPA framework (Definition 2):

- The numerator and denominator of the approximant are extended from **polynomial functions** $P(z) = \sum a_i z^i$ to **series composed of any continuous function**: $P(z) = \sum a_i \boldsymbol{g_i(z)}$ where $g_i : \mathbb{C} \to \mathbb{C}$ can be *any* continuous function.
- An **auxiliary operator** $\Gamma(z)$ (any differentiable function or constant) is introduced.
- The classical Padé and all known quasi-Padé variants (exponential, circular, etc.) become special cases of this definition.

This flexibility is key: different $g_i(z)$ can be chosen to match the geometric character of the solution being approximated—hyperbolic secant for homoclinic orbits (symmetric approach to a saddle), hyperbolic tangent for heteroclinic orbits (connection between two different saddles), and polynomial basis for intermediate-order corrections.

3 The GPLP Procedure (Four Steps)

The oscillator is expanded: $x = \sum \varepsilon^n x_n$, $\mu_c = \sum \varepsilon^n \mu_{cn}$, yielding a hierarchy of perturbation equations (21)–(24) in the original paper. The GPLP method solves each without integration:

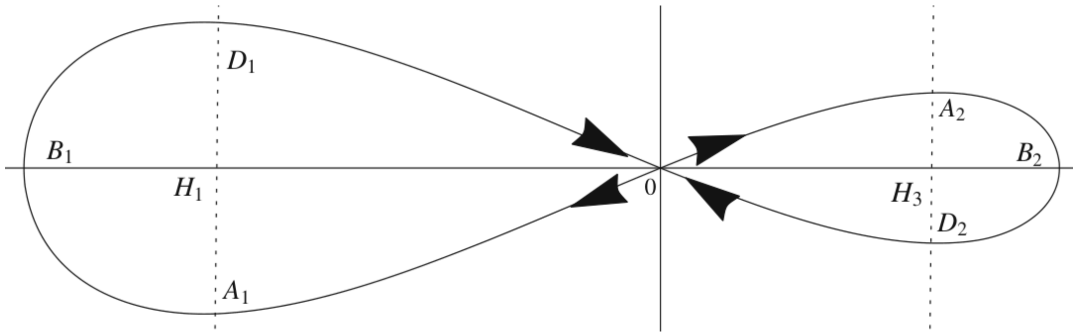

Figure 1: Potential energy curve and phase portraits of the Helmholtz–Duffing oscillator (66) at $\varepsilon = 0$. The cubic asymmetric potential yields two homoclinic orbits around the saddle point $H_2(0,0)$, symmetric about the x-axis but asymmetric about the $\dot{x}$-axis. Arrows: time evolution as $t \to \pm\infty$.

3.1 Step I: Zero-order x_0 (conservative orbit)

- Derive power series $x_0(t) = \sum a_i t^i$ from the conservative equation (21). Initial values: $a_1 = 0$, a_0 from $\int_{h_i}^{a_0} g(x)\mathrm{d}x = 0$.
- **Homoclinic:** construct $\mathrm{GPA}[L_0/M_0]x_0(t) = \frac{\sum \alpha_i \,\mathrm{Sech}^i t}{1+\sum \beta_i \,\mathrm{Sech}^i t}$ — Sech basis satisfies $x(\pm\infty) = h_i$, $\dot{x}(\pm\infty) = 0$.
- **Heteroclinic:** use $\mathrm{Tanh}^i t$ basis — satisfies $x(+\infty) = h_i$, $x(-\infty) = h_j$ $(i \neq j)$.
- Match Taylor coefficients to determine α_i, β_i.

3.2 Step II: First-order x_1 and μ_{c0}

- Expand $x_1(t) = \sum b_i t^i$, $b_0 = b_1 = 0$. Substitute into Eq. (22) to get $b_i(\mu_{c0})$.
- Construct $\mathrm{GPA}[L_1/M_1]x_1(t) = \mathrm{Tanh}(t)\frac{\sum \eta_i t^i}{1+\sum \xi_i t^i}$ — the Tanh prefactor enforces $x_1(\pm\infty) = 0$.
- $\lim_{t\to\pm\infty} x_1(t) = 0$ yields $\eta_{L_1}(\mu_{c0}) = 0$, giving μ_{c0} — the **zero-order critical bifurcation parameter**. This condition is shown equivalent to the classical Melnikov condition (Eq. 40–43).

3.3 Step III: Second-order x_2 and d_0/d_1

Homoclinic: $d_1 = 0$, $\mu_{c1} = 0$. Heteroclinic: $d_0 = 0$, $\mu_{c1} = 0$. Construct $\mathrm{GPA}[L_2/M_2]x_2(t) = \frac{\sum \lambda_i t^i}{1+\sum \nu_i t^i}$; $\lambda_{L_2} = 0$ determines d_0 or d_1.

3.4 Step IV: Third-order x_3 and μ_{c2}

$\mathrm{GPA}[L_3/M_3]x_3(t) = \frac{\sum \sigma_i t^i}{1+\sum \tau_i t^i}$ with $r_0 = r_1 = 0$. $\sigma_{L_3}(\mu_{c2}) = 0$ yields second-order correction.
Final result: $x(t) = \sum_{i=0}^{3} \varepsilon^i \,\mathrm{GPA}[L_i/M_i]x_i(t) + O(\varepsilon^4)$, $\quad \mu_c = \mu_{c0} + \varepsilon^2\mu_{c2} + O(\varepsilon^3)$.

4 Why Three Different Approximant Types?

A critical design choice in GPLP is using **different approximant forms** for different orders, rather than a single unified form throughout. The paper provides a detailed rationale (Remark 5, Example 5):

- x_0 uses Sech/Tanh basis because these **exactly respect the asymptotic boundary conditions** of H/H orbits. A polynomial-based GPA for x_0 would require many more terms to achieve comparable accuracy.

- x_1 uses a Tanh-prefixed polynomial form because the first-order correction x_1 must satisfy $x_1(\pm\infty) = 0$ — the Tanh prefactor enforces this automatically, while a pure polynomial would require solving an additional constraint equation.
- x_2, x_3 use polynomial-based GPA because by these orders the asymptotic conditions are already embedded in the lower-order solutions, and polynomial forms offer simpler coefficient matching.

The paper explicitly compares the mixed-form approach against using the same approximant throughout (Example 5), confirming that the tailored mixed-form strategy yields substantially higher accuracy.

5 Applications: Four Oscillator Classes

5.1 1. Generalized Helmholtz–Duffing–Van der Pol Oscillator

Governing equation:

$$\ddot{x} + c_1 x + c_2 x^2 + c_3 x^3 = \varepsilon(\mu_c + \mu_1 x + \mu_2 x^2)\dot{x}$$

Example 1 (Eq. 66): $c_1 = -2$, $c_2 = 5$, $c_3 = 5$, $\mu_1 = 1$, $\mu_2 = 1$. Saddle point at $H_2(0, 0)$, equilibrium at $H_1(-1.3062, 0)$ and $H_3(0.3062, 0)$. Two asymmetric homoclinic orbits coexist. Key results:

- **Zero-order GPA solution:**

$$\boxed{\mathrm{GPA}[4/4]x_0(t) = \frac{0.0814\,\mathrm{Sech}\,t + 3.2534\,\mathrm{Sech}^2 t + 16.0572\,\mathrm{Sech}^3 t + 16.8002\,\mathrm{Sech}^4 t}{1 + 14.2944\,\mathrm{Sech}\,t + 39.3405\,\mathrm{Sech}^2 t + 22.9087\,\mathrm{Sech}^3 t + 566.4437\,\mathrm{Sech}^4 t}}$$

- **Critical bifurcation parameter:** $\mu_{c0} \approx -6.3941$
- **Accuracy:** maintained at $\varepsilon = 2, 3$ (Fig. 2) and even at $\varepsilon = 50, 60$ for the SD oscillator variant (Fig. 4)

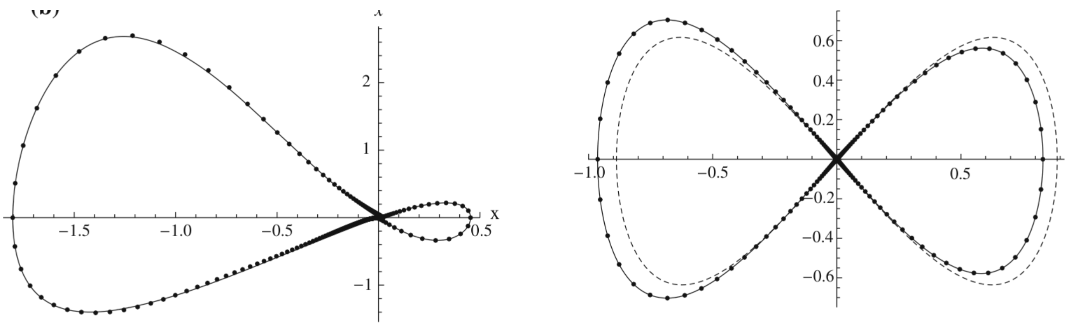

Figure 2: Homoclinic orbits of Eq. (66) for (a) $\varepsilon = 2$ and (b) $\varepsilon = 3$. Solid: Runge–Kutta. Dash-dot: GPLP (ε^3 expansion). Note that the ε^3 solution maintains accuracy even at moderately large perturbation parameters.

5.2 2. Duffing-Harmonic–Van der Pol Oscillator (Rational Restoring Force)

For $\ddot{x} + \frac{\lambda x + \eta x^3}{1 + \nu x^2} = \varepsilon(\mu_c + \mu_1 x + \mu_2 x^2)\dot{x}$ with $\lambda = -5$, $\eta = 5$, $\nu = 5$, $\mu_{c0} = -1.9026$. This demonstrates that GPLP handles rational restoring forces naturally — the only change is in the coefficients of $g(x)$ expansion.

5.3 3. SD Oscillator (Irrational Restoring Force)

For $\ddot{x} + x\left(1 - \frac{1}{\sqrt{x^2+\alpha^2}}\right) = \varepsilon(\mu_c + \mu_1 x + \mu_2 x^2)\dot{x}$, GPLP successfully predicts homoclinic orbits even at **extremely large** $\varepsilon = 50, 60$ (Fig. 4). This is possible because the zero-order GPA solution already closely approximates the exact conservative orbit, and the ε^3 expansion captures higher-order damping effects accurately.

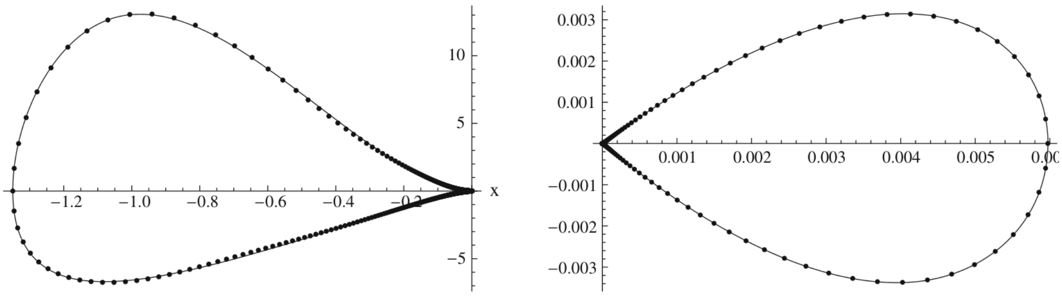

Figure 3: Homoclinic orbits of the SD oscillator at (a,b) $\varepsilon = 50$ and (c,d) $\varepsilon = 60$. Solid: Runge–Kutta. Dash-dot: GPLP. The ε^3 GPLP solution remains accurate at perturbation parameters two orders of magnitude larger than typical "weakly nonlinear" regimes.

5.4 4. Φ^6-Van der Pol Oscillator (Quintic, Simultaneous H/H)

For $\ddot{x} + c_1 x + c_3 x^3 + c_5 x^5 = \varepsilon(\mu_c + \mu_1 x + \mu_2 x^2)\dot{x}$ with appropriately chosen coefficients, the quintic potential creates a landscape where **homoclinic and heteroclinic orbits coexist at the same saddle**. GPLP predicts **both critical parameter values simultaneously** (Figs. 10–12 in the original paper).

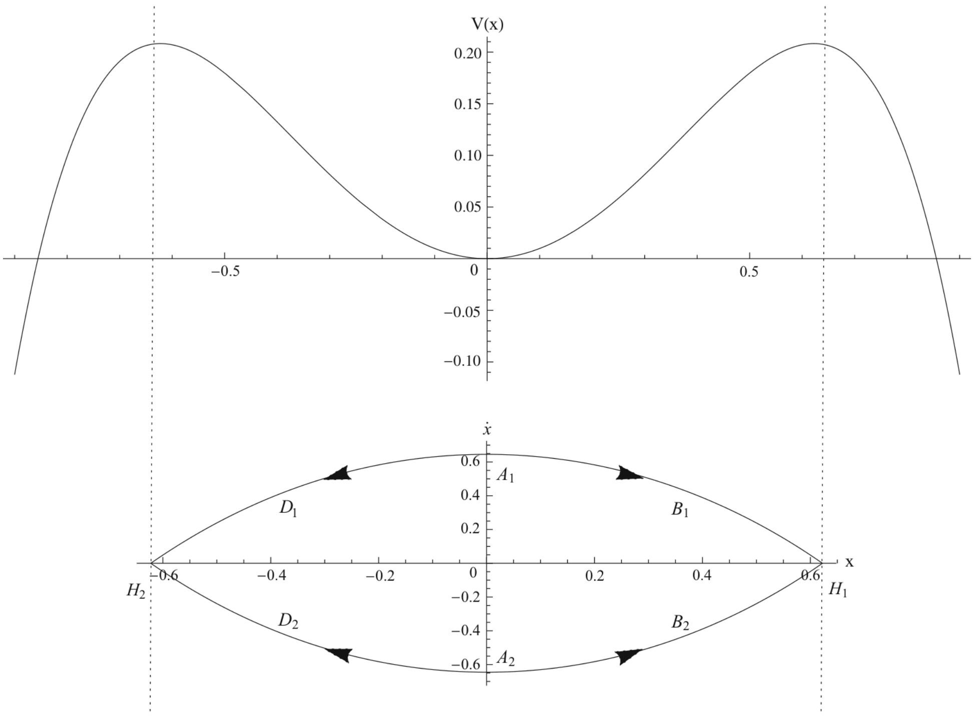

Figure 4: Potential energy curve and heteroclinic orbits of the Φ^6-Van der Pol oscillator (102) at $\varepsilon = 0$. The quintic potential supports simultaneous homoclinic and heteroclinic orbits.

5.5 SD Oscillator with Dual-Well Structure (Further Example)

The SD oscillator with parameters creating a double-well potential (Eq. 112) exhibits both homoclinic orbits at different saddle points. GPLP handles this multi-saddle configuration without additional modifications.

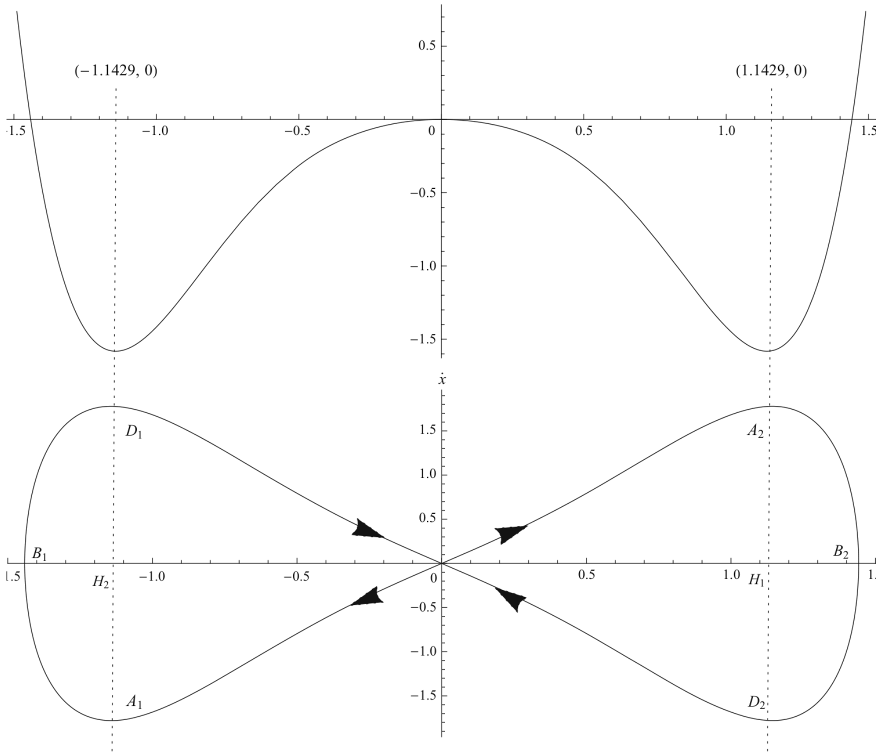

Figure 5: Potential energy curve and homoclinic orbits of the SD oscillator (112) with double-well structure. The two homoclinic orbits around different saddles are both accurately captured.

5.6 Complex Multi-Orbit System

The most general oscillator studied (Eq. 121) features a potential landscape where **homoclinic and heteroclinic orbits coexist at the same saddle** in a single system. The GPLP framework handles this maximally complex case without any procedural modification (Fig. 10).

6 Key Findings

1. **No-integration perturbation method:** GPLP replaces all integration—the computational bottleneck of traditional methods—with coefficient matching. High-order ε^3 solutions become accessible for complicated oscillators.

2. **Three tailored approximant types:** Sech-based (homoclinic), Tanh-based (heteroclinic), and polynomial-based (intermediate corrections) — each chosen to exploit the geometric character of the specific solution being approximated. The mixed-form strategy outperforms any single-form approach (Example 5).

3. **Universal applicability:** Validated on Helmholtz–Duffing (cubic), Φ^6 (quintic), Duffing-harmonic (rational), and SD (irrational from geometric constraint) — covering all canonical restoring-force types.

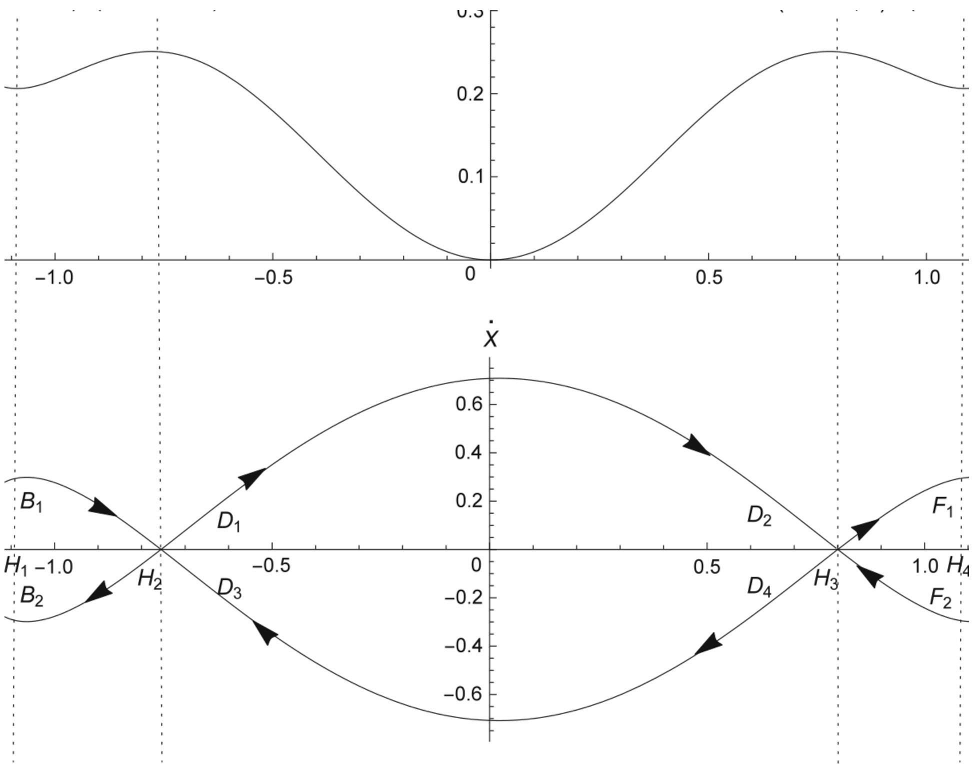

Figure 6: Potential energy curve and simultaneous homoclinic/heteroclinic orbits of Eq. (121) at $\varepsilon = 0$. This is the most complex system studied: multiple saddles, coexisting homoclinic and heteroclinic orbits, all captured in a unified GPLP framework.

4. **Accuracy at large ε:** ε^3 expansion gives accurate results at $\varepsilon = 50$–60 (SD oscillator, Fig. 4). This is possible because the zero-order GPA already closely matches the exact conservative orbit.

5. **Simultaneous H/H bifurcation prediction:** For oscillators where both homoclinic and heteroclinic orbits coexist (e.g., Φ^6-VDP, Eq. 121), both critical parameters are predicted analytically in one framework.

6. **Melnikov consistency:** The GPLP condition $\eta_{L_1}(\mu_{c0}) = 0$ is proved equivalent to the classical Melnikov condition, providing rigorous theoretical justification.

7 Method Limitations (Honest Assessment)

- System parameters must be set before calculation (unlike algebraic methods such as GHFP). Not ideal for tracing continuous parameter–solution dependencies.
- The approximation orders L_i, M_i must be manually chosen based on desired accuracy.
- Derivations (Eq. 20) become algebraically heavy at higher orders, even though integration is avoided.

Author: Zhenbo Li (lizhenbo@usc.edu.cn)
University of South China, School of Mathematics and Physics